

General and self-consistent method for the calibration of polarization modulators, polarimeters, and Mueller-matrix ellipsometers

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Calibration of polarization-state generators (PSG's), polarimeters, and Mueller-matrix ellipsometers (MME's) is an important factor in the practical use of these instruments. A new general procedure, the eigenvalue calibration method (ECM), is presented. It can calibrate any complete MME consisting of a PSG and a polarimeter that generate and measure, respectively, all the states of polarization of light. In the ECM, the PSG and the polarimeter are described by two 4×4 matrices **W** and **A**, and their 32 coefficients are determined from three or four measurements performed on reference samples. Those references are smooth isotropic samples and perfect linear polarizers. Their optical characteristics are unambiguously determined during the calibration from the eigenvalues of the measured matrices. The ECM does not require accurate alignment of the various optical elements and does not involve any first-order approximation. The ECM also displays an efficient error control capability that can be used to improve the MME behavior. The ECM is illustrated by an experimental calibration, at two wavelengths (458 and 633 nm), of a MME consisting of a coupled phase modulator associated with a prism division-of-amplitude polarimeter. © 1999 Optical Society of America

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1. Introduction

A comprehensive polarimetric characterization of a sample is usually represented by the sample's 4×4 real Mueller matrix. This matrix describes the modification of polarization of light after interaction with a sample in the general linear case. Indeed, the use of the Mueller-matrix formalism becomes necessary whenever the sample is partially or totally depolarizing, as in cases of rough surfaces and scattering media. Mueller-matrix ellipsometry can be considered an extension of the classic and generalized ellipsometry that is restricted to nondepolarizing samples.

It is important, when one is using a Mueller-matrix ellipsometer (MME), to calibrate accurately the two optical arms that consist of a complete polarization-

state generator (PSG) and a complete polarimeter.¹ As defined in Ref. 1, "complete" means that the PSG and the polarimeter can generate and measure the four states of polarization of light.

Usually the calibration of a polarimetric tool is a two-step process. First, one precisely orients the optical elements, a procedure that requires high-precision devices together with careful alignment. In a second stage, the behavior of the optical elements must conveniently be described and characterized by physical properties. The behavior can be distorted by first-order failures, as in the case of a photoelastic modulator.² Thus one has to combine several measurements in different configurations to determine the main values and the shortcomings. Those measurements are adapted to each optical setup. These aspects of the calibration process have already been reviewed in the literature.³⁻⁵ Furthermore, one can notice that usual calibration procedures are difficult to implement *in situ* because of the influence of the optical elements that are necessarily included in the light path (filters, windows, lenses, mirrors, etc.). All these elements can induce modifications of the state of polarization of light.

The eigenvalue calibration method (ECM) consists first in extending the use of the matrix formalism to the global experimental setup to be calibrated.

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Three characteristic 4×4 real matrices are naturally introduced: $\mathbf{W}$ for the entrance arm (PSG), $\mathbf{A}$ for the exit arm (polarimeter), and $\mathbf{M}$, the Mueller matrix of the sample. Then a basic measurement is the matrix product $\mathbf{AMW}$, i.e., 16 coefficients. The ECM takes advantage of this matrix representation in making extensive use of linear algebra. With the ECM the entire calibration of the 32 coefficients of a MME ($\mathbf{A}$ and $\mathbf{W}$) can be performed with only three or four measurements of reference samples. The reference samples used here are smooth, isotropic surfaces and linear polarizers.

The ECM has three advantages: First, no assumption has to be made about the system that is being calibrated, except that it must be complete. All the optical elements are automatically included in the matrix representation, even if their behavior is not completely described; this enables the existing MME to be simplified because the orientations and positions of the various elements do not have to be precisely adjusted. For example, the presence of a stressed window will automatically be taken into account when the ECM is used. The procedure is also universal because it depends only on the choice of the reference samples; first-order approximations are not necessary. Second, the few characteristics of the reference samples are completely determined during the calibration, without the need for subsidiary measurements: the Fresnel reflection coefficients and the polarizer orientation and transmission factors are deduced from the eigenvalues of the measured matrices. If a spectroscopic MME is calibrated by the ECM, $\mathbf{W}$ and $\mathbf{A}$ will depend on the wavelength λ , and the characteristics of the reference samples will be determined for each wavelength in the same manner. Third, the accuracy of the calibration procedure can be precisely evaluated when the ECM is used, as described below.

After a brief description of the Mueller-matrix formalism we review the ways in which the various MME configurations can easily be described with a matrix formalism. Then we describe the ECM in detail, beginning with the simplest configuration, in which the MME can produce measurements in a reflection-and-transmission configuration. The other practical situation, in which the MME can produce measurements only in a reflection configuration for a limited number of angles of incidence, is deduced directly from the first case. In the last part of the paper we display experimental results obtained during the calibration of a multiwavelength MME consisting of a coupled phase modulator⁶ associated with a prism division-of-amplitude polarimeter⁷ at 458 and 633 nm.

2. Matrix Formalism for Polarimeters, Polarization-State Generators, and Mueller-Matrix Ellipsometers

A. Mueller Matrices: Reference Samples

Throughout this paper the more general state of polarization of a light beam is represented by its Stokes vector⁸⁻¹⁰ $\mathbf{S} = {}^T(I, Q, U, V)$, where T is the transpo-

sition operator. The linear interaction with a sample is described by the four-by-four real Mueller matrix of the sample. As a consequence of the linearity, the overall matrix $\mathbf{M}$ of an optical system is obtained from the matrix product of its components $\mathbf{M}_i$:

$$\mathbf{M} = \mathbf{M}_n \mathbf{M}_{n-1} \dots \mathbf{M}_2 \mathbf{M}_1. \quad (1)$$

Only two types of Mueller matrix that have perfectly known theoretical forms are used in this paper; they are called the reference samples. The first ones are the linear polarizer matrices $\mathbf{P}(\tau, \theta)$, where θ designates the orientation of the polarizer and τ is the transmission coefficient:

$$\mathbf{P}(\tau, 0) = \frac{\tau}{2} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$\mathbf{P}(\tau, \pi/4) = \frac{\tau}{2} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad (2)$$

The Mueller matrices with other orientations are obtained by use of the rotation

$$\mathbf{P}(\tau, \theta) = \mathbf{U}(\theta) \mathbf{P}(\tau, 0) \mathbf{U}(-\theta), \quad (3)$$

where

$$\mathbf{U}(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\theta & -\sin 2\theta & 0 \\ 0 & \sin 2\theta & \cos 2\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (4)$$

In this paper, polarizer matrices need to be perfect. This requirement leads to the use of crystal polarizers (Rochon, Glan, Wollaston, etc.) in the near-ultraviolet–near infrared range, which have an excellent extinction ratio that is less than 10^{-5} . Dichroic sheet polarizers for near-ultraviolet–near-infrared or grid polarizers for infrared cannot be represented in this way with satisfactory accuracy.

The second class of theoretically well-known systems corresponds to smooth isotropic samples. The reflection on their surface can be decomposed into a partially polarizing effect (Ψ) and a phase-shifting effect (Δ), where Ψ and Δ correspond to the usual ellipsometric angles related to the Fresnel coefficients [$\tan(\Psi)\exp(i\Delta) = r_p/r_s$ in reflection]. The reflection Mueller matrix $\mathbf{R}(\tau, \Psi, \Delta)$ of such a sample is

[where $\tau = 0.5(|r_p|^2 + |r_s|^2)$ is the reflection coefficient for unpolarized light]

$$\mathbf{R}(\tau, \Psi, \Delta) = \tau \times \begin{bmatrix} 1 & -\cos 2\Psi & 0 & 0 \\ -\cos 2\Psi & 1 & 0 & 0 \\ 0 & 0 & \sin 2\Psi \cos \Delta & \sin 2\Psi \sin \Delta \\ 0 & 0 & -\sin 2\Psi \sin \Delta & \sin 2\Psi \cos \Delta \end{bmatrix}. \quad (5)$$

B. Polarization-State Generator, Polarimeter, and Mueller-Matrix Ellipsometer

To give a general presentation of the ECM, we present a short, nonexhaustive review of the various techniques used in the PSG–polarimeter setups. The objective is to get a unified formalism in which for all the setups the PSG and the polarimeter are represented by two characteristic 4×4 real matrices $\mathbf{W}$ and $\mathbf{A}$. The calibration procedure will finally consist of determining their 2×16 coefficients.

1. Polarization-State Generator

A PSG is a light source that is able to generate different states of polarization. Only complete PSG's that can generate at least four independent states of polarization are considered in this paper. Their matrices $\mathbf{W}$ are necessarily invertible.

The simplest nonmodulated manual PSG consists of various polarizing elements located successively in a light beam to generate a set of Stokes vectors. These Stokes vectors $\mathbf{S}_i$ can be written as the product of the PSG matrix $\mathbf{W}$ and at least four input vectors $\mathbf{e}_i$:

$$\mathbf{S}_i = \mathbf{W}\mathbf{e}_i. \quad (6)$$

For example,¹¹ using a polarizer successively at 0, 45°, and 90° and a polarizer at 0° followed by a quarter-wave plate at 45° leads to (where τ_i is the coefficient of transmission)

$$\mathbf{W} = \frac{1}{2} \begin{bmatrix} \tau_1 & \tau_1 & \tau_1 & \tau_2 \\ \tau_1 & 0 & -\tau_1 & 0 \\ 0 & \tau_1 & 0 & 0 \\ 0 & 0 & 0 & \tau_2 \end{bmatrix}, \quad \mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \dots, \quad \mathbf{e}_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad (7)$$

In the case of a time-modulated PSG there is only one input vector, consisting of a set of harmonic functions:

$$\mathbf{S}(t) = \mathbf{W}\mathbf{e}(t). \quad (8)$$

Harmonic vector $\mathbf{e}(t)$ depends on the modulation technique.^{5,6,12–14} Some examples are given in Table 1. A review paper¹ describes most of these examples.

Table 1. Composition of the Harmonic Vector for Four Modulation Techniques

Low-frequency modulation
Fixed polarizer and rotating compensator ^a
$\mathbf{e}(t) = {}^T(1, \sin 2\omega t, \cos 4\omega t, \sin 4\omega t)$
Element axis, ωt
Fixed polarizer and oscillating compensator ^b
$\mathbf{e}(t) = {}^T(1, \sin \omega t, \cos 2\omega t, \sin 3\omega t)$
Element axis, $c_0 + c_1 \sin(\omega t)$
High-frequency modulation
Dual phase modulator ^c
$\mathbf{e}(t) = {}^T[1, \cos 2\omega_2 t, \cos(\omega_1 - \omega_2)t, \sin(2\omega_1 - \omega_2)t]$
Two phase modulations $\delta_{1,2}(t) = A_{1,2} \sin \omega_{1,2}t$
Coupled phase modulator ^d
$\mathbf{e}(t) = {}^T(\sin \omega t, \cos 2\omega t, \sin 3\omega t, \cos 4\omega t)$
One phase modulation $\delta(t) = A \sin(\omega t)$

^aRef. 16.

^bRef. 13.

^cRef. 5.

^dRef. 6 and 14.

2. Polarimeter

A polarimeter is dedicated to the measurement of an unknown Stokes vector. It can be described by a characteristic matrix $\mathbf{A}$ that links the measured intensity(ies) to the input Stokes vector. Only complete polarimeters, which are able to measure all the states of polarization, are considered here. Their matrices $\mathbf{A}$ are necessarily invertible.

All the previous PSG setups can be used as polarimeters. In those cases, the direction of light is inverted: matrix $\mathbf{A}$ instead of $\mathbf{W}$, output vectors $\mathbf{f}$ instead of input vectors $\mathbf{e}$, and a photodetector instead of a light source are used. The detected intensity is then related to Stokes vector $\mathbf{S}$ by measurement

$$I_i = {}^T\mathbf{f}_i\mathbf{A}\mathbf{S} \quad (i = 1, \dots, 4),$$

or

$$I(t) = {}^T\mathbf{f}(t)\mathbf{A}\mathbf{S}. \quad (9)$$

Another class of polarimeter that is different from those used in the former PSG techniques has been described in the literature. It essentially includes a four-detector photopolarimeter¹⁵ and a division-of-amplitude polarimeter.^{7,16–18} These polarimeters use four detectors instead of one, so four intensities I_{1-4} can be measured simultaneously. The Stokes vector is then directly related to these four intensities by

$$\begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{pmatrix} = \mathbf{A}\mathbf{S}. \quad (10)$$

3. Mueller-Matrix Ellipsometer

A MME is composed of a complete PSG and a complete polarimeter. By combining measurements for the sets of input and output vectors, the basic capacity of any MME is to measure the matrix product

AMW, where **M** is the Mueller matrix of the sample under investigation. The experimental procedure and treatment of data depend on the PSG and the polarimeter that are used. The simplest MME uses a nonmodulated PSG and polarimeter.¹⁹ **AMW** is obtained from the 16 intensities I_{ij} associated with input and output vectors i and j :

$$\mathbf{AMW} = [I_{ij}]_{i,j=1..4}, \quad (11)$$

where

$$I_{ij} = {}^t\mathbf{f}_i \mathbf{AMW} \mathbf{e}_j.$$

With a modulated PSG and polarimeter **AMW** is obtained from the Fourier analysis of a single time-varying signal¹²:

$$I(t) = {}^t\mathbf{f}(t) \mathbf{AMW} \mathbf{e}(t). \quad (12)$$

When a four-channel polarimeter is used, **AMW** is obtained from the measurement of four intensities^{20,21}:

$$\begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{pmatrix}(t) = \mathbf{AMW} \mathbf{e}(t)$$

or

$$\begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{pmatrix}_i = \mathbf{AMW} \mathbf{e}_i. \quad (13)$$

The configurations described above are of course nonunique, and other arrangements can be proposed or found in the literature. The configurations that use a manual setup are often used to calibrate a polarimeter or a PSG, and two matrices are generally involved, even if only one is needed for calibration.

Before going into the details of the calibration procedure we point out that, although the matrices **A** and **W** are most of the time 4×4 real matrices, the general form is $n \times 4$ for **A** and $4 \times m$ for **W**, where n and m are greater than 4. $n > 4$ means that more than four input Stokes vectors are generated, and $m > 4$ means that the output Stokes vectors are overdetermined. For example, the 16-beam grating-based division-of-amplitude polarimeter described recently²² has a 4×16 characteristic matrix. In this paper the indicated dimensions refer to 4×4 **A** and **W** matrices; nevertheless the ECM is fully compatible with other sizes, except that the dimensions of the intermediate matrices should be adapted.

3. Calibration Theory

We assume at first that the MME system provides the measurement of the matrix product **AMW** for any Mueller matrix, including the identity matrix. It is assumed that the MME can work in both the reflection and the direct-transmission configurations, as displayed in Fig. 1. It is also assumed that the

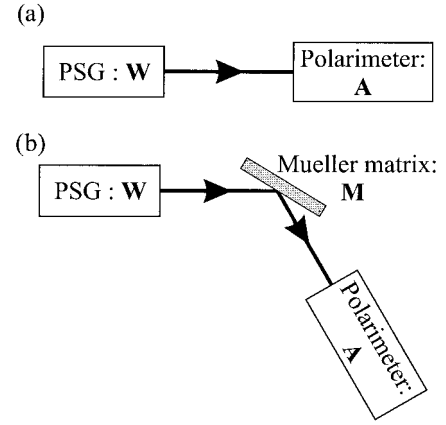


Fig. 1. Measurement configurations involved in the ECM. (a) Direct transmission measurement without a sample: The corresponding measurement is **aw**. (b) The reflection configuration: the corresponding measurement is **amw**.

Mueller matrices **M** of the reference samples are perfectly known. In what follows, we describe in detail how to deal with these limitations.

The following conventions are used in this paper: capital-letter matrices (such as **M** and **X**) refer to theoretical matrices, whereas boldface italic lower-case letters, such as (**amw**), correspond to the experimental measurements. Without experimental errors, the 4×4 measured matrix (**amw**) is equal to the matrix product **AMW**. If **M** is known, this equality provides the basic information used by the ECM to determine the matrices **A** and **W**.

With respect to the matrix algebra, mathematical support can be found in many textbooks,^{23,24} and one should notice that the matrices **A**, **W**, and **R**(τ, Ψ, Δ) defined in Eq. (5) are invertible, **A** and **W** because the PSG and the polarimeter are complete and **R**(τ, Ψ, Δ) because Ψ is assumed to be different from zero and $\pi/2$. Consequently, their measured products (**aw**) and (**arw**) are also invertible matrices.

A. Mathematical Construction

1. System of Equations

We define a linear mapping $\mathbb{H}_M$ from the set $\mathbb{M}_4(\mathbb{R})$ of the real 4×4 matrices into itself. It depends on the parameter **M**, the Mueller matrix of a reference optical element, on (**amw**), the corresponding measurement, and on (**aw**), the measurement in direct transmission without samples:

$$\begin{aligned} \mathbb{H}_M: \mathbb{M}_4(\mathbb{R}) &\rightarrow \mathbb{M}_4(\mathbb{R}), \\ \mathbf{X} &\rightarrow \mathbf{MX} - \mathbf{X}(\mathbf{aw})^{-1}(\mathbf{amw}). \end{aligned} \quad (14)$$

$\mathbb{H}_M$ has the property of containing **W** within its null space [$\mathbb{H}_M(\mathbf{W}) = 0$], whatever the value of **M**, because without experimental errors (**aw**)⁻¹(**amw**) is equal to **W**⁻¹**MW**. Moreover, inasmuch as **W** belongs to the null space of all applications $\mathbb{H}_M$, it also belongs to their intersection. Finally, a well-chosen set of reference samples $\{\mathbf{M}_1, \dots, \mathbf{M}_n\}$, as described in Subsection 3.B below, can reduce the number of solutions

to one: $\mathbf{W}$ is determined as being the unique solution of

$$\mathbb{H}_M(\mathbf{X}) = 0, \quad \mathbf{M} \in \{\mathbf{M}_1, \dots, \mathbf{M}_n\}. \quad (15)$$

$\mathbf{A}$ is then obtained from the measured matrix product $(\mathbf{aw})$:

$$\mathbf{A} = (\mathbf{aw})\mathbf{W}^{-1}. \quad (16)$$

2. Solution of the System

Assuming that an appropriate set of reference samples $\{\mathbf{M}\} = \{\mathbf{M}_1, \dots, \mathbf{M}_n\}$ has been chosen, it is useful to adopt a matrix formulation to solve the linear system of Eq. (15). As a matter of fact, a 4×4 matrix $\mathbf{X}$ can be written as a vector $\mathbf{X}^{16} = (x_i)_{i=1}^{16}$ in an appropriate basis set $\{\mathbf{g}_i\}_{i=1, \dots, 16}$ of $\mathbb{M}_4(\mathbb{R})$ such that the 4×4 matrix $\mathbf{X}$ is equal to $\sum_{i=1, \dots, 16} x_i \cdot \mathbf{g}_i$. On this basis, the linear mapping $\mathbb{H}_M$ is represented by a 16×16 real matrix $\mathbf{H}_M$, the 4×4 matrix $\mathbb{H}_M(\mathbf{X})$ is equivalent to the product $\mathbf{H}_M \mathbf{X}^{16}$, and $\mathbf{W}$, the solution of Eq. (15), is represented by the vector $\mathbf{W}^{16}$ solution of

$$\mathbf{H}_M \mathbf{X}^{16} = 0, \quad \mathbf{M} \in \{\mathbf{M}_1, \dots, \mathbf{M}_n\}. \quad (17)$$

The solution of the overdetermined linear system (17) by a least-squares method is given by the well-known relation²⁴

$$\mathbf{K} \mathbf{X}^{16} = 0, \quad (18)$$

where

$$\mathbf{K} = \begin{pmatrix} \mathbf{H}_{M1} & \mathbf{H}_{M1} \\ \vdots & \vdots \\ \mathbf{H}_{Mn} & \mathbf{H}_{Mn} \end{pmatrix}^T = {}^T \mathbf{H}_{M1} \mathbf{H}_{M1} + \dots + {}^T \mathbf{H}_{Mn} \mathbf{H}_{Mn}. \quad (19)$$

The matrix $\mathbf{K}$ is a positive symmetric real matrix, so it can be diagonalized. It has 15 non-null and 1 null eigenvalue because Eq. (18) has only one solution, $\mathbf{W}^{16}$, which is the unique eigenvector associated with the null eigenvalue.

We have found the solution to the calibration problem from a mathematical point of view but without taking into account the physical limitations: Practically, the equality $\mathbb{H}_M(\mathbf{W}) = 0$ can never be exactly verified. Because of the limited experimental precision, $\mathbf{W}(\mathbf{aw})^{-1}(\mathbf{aw})$ is always slightly different from $\mathbf{MW}$. This fact imparts peculiar properties to eigenvalues $(\lambda_1 \dots \lambda_{16})$ of $\mathbf{K}$. They are all different from zero except λ_{16} , the smallest, which should theoretically be null and which is close to zero in reality. One gets

$$\mathbf{K} = {}^T \mathbf{O} \begin{bmatrix} \lambda_1 & & 0 & \cdots & 0 \\ & \ddots & & & \vdots \\ 0 & & \lambda_i & & 0 \\ \vdots & & & \ddots & \\ 0 & \cdots & 0 & & \lambda_{16} \end{bmatrix} \mathbf{O}, \quad \lambda_1 > \dots > \lambda_{15} \gg \lambda_{16} \approx 0, \quad (20)$$

where $\mathbf{O}$ is the orthonormal matrix whose rows are the eigenvectors of $\mathbf{K}$. The null space of $\mathbf{K}$ is empty

in practical circumstances, so the solution $\mathbf{W}^{16}$ is the eigenvector of $\mathbf{K}$ or the row of $\mathbf{O}$ associated with the smallest eigenvalue:

$$\mathbf{W}^{16} = {}^T \mathbf{O} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}. \quad (21)$$

Finally, $\mathbf{W}^{16}$ is written back as a 4×4 matrix $\mathbf{W}$ and $\mathbf{A}$ is obtained with Eq. (16) from the measured matrix product $(\mathbf{aw})$. The values of $\mathbf{W}$ and $\mathbf{A}$ have been fully determined, apart from a multiplication factor α that is not important because $\mathbf{W}$ and $\mathbf{A}$ can be changed into $\alpha \mathbf{W}$ and $\alpha^{-1} \mathbf{A}$ without consequences.

B. Choice of the Reference Samples Set $\{\mathbf{M}\}$

1. General Properties

In practice, the samples used are restricted to those whose theoretical Mueller matrices have a perfectly well-known structure. These matrices are functions of intrinsic parameters such as the transmission or partial polarizing coefficients defined in Eqs. (2) and (5). Their choice is ruled by the necessity to reduce the intersection of the null spaces of the mappings $\mathbb{H}_M$, defined in Eq. (14), to get only one solution, which determines $\mathbf{W}$.

The first important and interesting property, demonstrated in Appendix A, is that the choice of reference samples is universal: It does not depend on $\mathbf{W}$ and $\mathbf{A}$. This consequently emphasizes the utility of the present calibration method, which can address any kind of technology within the limitation that the PSG and the polarimeter be complete.

In Appendix B we demonstrate that the number of matrices that commute with the reference matrices depends only on the multiplicity of eigenvalues of the matrices, as displayed in Eq. (B2) below. A direct consequence of Eq. (B2) is that using a single reference sample yields at least four solutions. Consequently, to yield only one solution ($\mathbf{W}$), the set of reference samples must include at least two matrices. Equation (B2) and expressions (B4) below demonstrate that using a reflection sample $\mathbf{R}(\tau, \Psi, \Delta)$ alone restricts the number of solutions of Eq. (15) to four when $\Psi \neq 0 [\pi/4]$ and $\Delta \neq 0 [\pi]$. In the same manner, a consequence of Eq. (B2) and expressions (B3) is that using a polarizer alone yields ten solutions. Finally, one can verify that the combination of a polarizer (with $\theta \neq 0 [\pi/2]$) and a reflection sample (with $\Psi \neq 0 [\pi/4]$ and $\Delta \neq 0 [\pi]$) enables us to get only one solution. The restriction on the polarizer angle is to avoid commutation with $\mathbf{R}(\tau, \Psi, \Delta)$, which would not reduce the number of solutions:

$\mathbf{W}$ is the unique solution of $\mathbb{H}_M(x) = 0$,

$$\mathbf{M} \in \{\mathbf{M}\} \text{ when } \{\mathbf{M}\} = \{\mathbf{R}(\tau_R, \Psi, \Delta), \mathbf{P}(\tau_P, \theta)\}. \quad (22)$$

2. Choice of the Characteristics of the Reference Samples

We have seen from expression (22) that the use of a set of reference samples including at least a reflection on a smooth sample and a polarizer, properly oriented, completely determines $\mathbf{W}$ and $\mathbf{A}$. Nevertheless, one has to choose among the characteristics of $\mathbf{R}(\tau_R, \Psi, \Delta)$ and $\mathbf{P}(\tau_P, \theta)$ those that will enable $\mathbf{W}$ to be determined with the best precision. A theoretical study of propagation of the experimental errors is beyond the scope of this paper. Nevertheless, it is demonstrated in Appendix C that it is necessary to ensure a good balance among the 15 nonzero eigenvalues of the matrix $\mathbf{K}$. The eigenvalues of $\mathbf{K}$ being sorted, as in Eq. (20),

the accuracy is maximum when λ_{15}/λ_1 is maximum.

$$(23)$$

The conditioning of $\mathbf{K}$, as defined in expression (23), is of course affected by $\mathbf{A}$ and $\mathbf{W}$, and the calibration is less accurate if $\mathbf{A}$ and $\mathbf{W}$ are ill-conditioned matrices (close to noninvertible matrices, i.e., the norms of their eigenvalues are not of the same order of magnitude). This means that MME's that provide equal sensitivity to all the Mueller matrices can be calibrated with higher accuracy.

A numerical procedure that uses Powell's method based on a steepest-descent algorithm²⁴ and optimizes criterion (23) provides the optimum values for the characteristics of the reference samples. When only two reference samples, $\mathbf{R}(\tau_R, \Psi, \Delta)$ and $\mathbf{P}(\tau_P, \theta)$, are used, with well-balanced matrices $\mathbf{W}$ and $\mathbf{A}$ one gets

$$\{\mathbf{M}\} = \{\mathbf{R}(\tau_R, \Psi, \Delta), \mathbf{P}(\tau_P, \theta)\}$$

is optimum when $\tau_R = 0.71$, $\Psi = \pi/6$ or $3\pi/6$,

$$\theta = \pm \pi/4, \quad \Delta = \pm \pi/2. \quad (24)$$

The physical systems that provide such values are discussed in Subsection 3.C./

3. Other Sets of Reference Samples

Values other than those displayed in Eq. (24) could be obtained if more than the two necessary matrices were used for calibration. In that case the precision of the calibration would increase because of the increase in the overdetermination. Other sets of reference samples also provide the needed properties. A particular set is worth describing because it enables us to calibrate a transmission-limited configuration (see Fig. 1). This set includes a phase shifter (a quarter-wave plate is optimum but not necessary) and a polarizer measured at 0° and 45° .

C. Experimental Determination of the Characteristics of the Reference Samples

In the theoretical discussion above we assumed a knowledge of the matrices of the reference samples. One could consequently argue that we have until now merely shifted the problem from the determination of

$\mathbf{W}$ and $\mathbf{A}$ to the determination of the characteristics of the reference samples. However, the advantage is that the indetermination relies no longer on a complex system but on well-known matrices that depend on a few parameters. Moreover, these parameters can be experimentally determined because the eigenvalues of a matrix product are independent of the product order. Consequently one can identify the theoretical eigenvalues of $\mathbf{M}$ to the eigenvalues of the product of measured matrices $(\mathbf{aw})^{-1}(\mathbf{amw})$. Identifying the eigenvalues of $\mathbf{R}(\tau_R, \Psi, \Delta)$ and $\mathbf{P}(\tau_P, \theta)$, displayed in expressions (B3) and (B4) below to be $(\lambda_{r1}, \lambda_{r2}, \lambda_{c1}, \lambda_{c2})$, the real and complex eigenvalues of $(\mathbf{aw})^{-1}(\mathbf{arw})$, leads to

$$\tau_p = \text{trace} [(\mathbf{aw})^{-1}(\mathbf{apw})], \quad \tau_r = 0.5(\lambda_{r1} + \lambda_{r2}),$$

$$\Delta = 0.5 \arg(\lambda_{c1}/\lambda_{c2}), \quad \Psi = \arctan \sqrt{\lambda_{r1}/\lambda_{r2}}, \quad (25)$$

where the trace is equal to the sum of the eigenvalues.

Note that in identifying the two sets of eigenvalues one cannot distinguish λ_{r1} from λ_{r2} and λ_{c1} from λ_{c2} , which results in an indetermination of Δ and Ψ values because permuting λ_{c1} and λ_{c2} changes the sign of Δ and permuting λ_{r1} and λ_{r2} changes Ψ into $\pi/2 - \Psi$. These indeterminations are of course easily removed merely by comparison of the obtained values with the theoretical ones.

The orientation θ of the polarizer cannot be recovered by identification of the eigenvalues because those of $\mathbf{P}(\tau_P, \theta)$ displayed in expressions (B3) do not depend on θ . A more physical approach consists in noticing that the orientation of the polarizer itself is meaningless if it is not related to the reflection sample, which is the basic reference and does not appear in $(\mathbf{aw})^{-1} \cdot (\mathbf{apw})$. Nevertheless, as soon as the other characteristics have been determined from the eigenvalues, θ can be obtained from matrix $\mathbf{K}$, defined in Eq. (20), which is a function of θ as follows:

$$\mathbf{K}(\theta) = {}^T \begin{pmatrix} \mathbf{H}_R \\ \mathbf{H}_{P(\theta)} \end{pmatrix} \begin{pmatrix} \mathbf{H}_R \\ \mathbf{H}_{P(\theta)} \end{pmatrix} = {}^T \mathbf{O}(\theta) \begin{bmatrix} \lambda_1(\theta) & & 0 \\ & \ddots & \\ 0 & & \lambda_{16}(\theta) \end{bmatrix} \mathbf{O}(\theta). \quad (26)$$

If θ has the correct value, all the eigenvalues of $\mathbf{K}$ are nonzero except one, as displayed in Eq. (20). On the other hand, when we depart from the actual value, the null space of $\mathbf{K}$ is empty and the system has only a trivial solution, which is the null matrix because all the eigenvalues are nonzero. As a consequence, we obtain the value of θ by minimizing the amplitude of the smallest eigenvalue of $\mathbf{K}$ with respect to the 15 other eigenvalues:

$$\theta \text{ verifies that } \frac{\lambda_{16}(\theta)}{\lambda_{15}(\theta)} \rightarrow 0. \quad (27)$$

The polarizer is then automatically referenced to the reflection sample.

D. Adaptation to the Reflection Configuration

We performed the operations above by assuming that the experimental setup was compatible with reflection and direct-transmission measurements (Fig. 1), which is of course not always the case. *In situ* MME's, for example, are often limited to the reflection configuration with a narrow angular range. On the other hand, polarimetric tools and especially polarimeters are often highly sensitive to the angle of incidence. Therefore, combining measurements in several configurations could generate experimental errors because of changes that would occur in matrix $\mathbf{A}$. Consequently it is worth describing how the calibration theory developed above can be adapted to a reflection-limited configuration.

When the MME operates in a diffraction-limited configuration, four measurements are required for calibrating $\mathbf{A}$ and $\mathbf{W}$ instead of three: two reflections, $(\mathbf{a}\mathbf{r}_1\mathbf{w})$ and $(\mathbf{a}\mathbf{r}_2\mathbf{w})$ with two different samples $\mathbf{R}_1$ and $\mathbf{R}_2$; and two others, including a polarizer after $(\mathbf{a}\mathbf{p}_a\mathbf{r}_1\mathbf{w})$ and before $(\mathbf{a}\mathbf{r}_1\mathbf{p}_b\mathbf{w})$, the reflection on $\mathbf{R}_1$. $\mathbf{W}$ is then determined in the same manner in the intersection of the null spaces of $\mathbb{H}_{\mathbf{P}_b}$ and $\mathbb{H}_{\mathbf{R}_{eq}}$:

$$\begin{aligned}\mathbb{H}_{\mathbf{P}_b}:\mathbf{X} &\rightarrow \mathbf{P}_b\mathbf{X} - \mathbf{X}(\mathbf{a}\mathbf{r}_1\mathbf{w})^{-1}(\mathbf{a}\mathbf{r}_1\mathbf{p}_b\mathbf{w}), \\ \mathbb{H}_{\mathbf{R}_{eq}}:\mathbf{X} &\rightarrow \mathbf{R}_{eq}\mathbf{X} - \mathbf{X}(\mathbf{a}\mathbf{r}_1\mathbf{w})^{-1}(\mathbf{a}\mathbf{r}_2\mathbf{w}),\end{aligned}\quad (28)$$

where $\mathbf{P}_b(\tau_b, \theta_b)$ is a polarizer matrix as in the previous case and $\mathbf{R}_{eq}$ is obtained from the reflection matrices $\mathbf{R}_1$ and $\mathbf{R}_2$:

$$\mathbf{R}_{eq}(\tau_{eq}, \Psi_{eq}, \Delta_{eq}) = \mathbf{R}_1^{-1}(\tau_1, \Psi_1, \Delta_1)\mathbf{R}_2(\tau_2, \Psi_2, \Delta_2), \quad (29)$$

with

$$\begin{aligned}\tau_{eq} &= \frac{\tau_2}{\tau_1} \frac{1 - \cos 2\Psi_1 \cos 2\Psi_2}{\sin^2 2\Psi_1}, \\ \tan \Psi_{eq} &= \cot \Psi_1 \tan \Psi_2, \\ \Delta_{eq} &= \Delta_2 - \Delta_1.\end{aligned}$$

Matrix $\mathbf{R}_{eq}$ has no physical meaning but has the structure of a reflection matrix described by Eq. (5). Finally, the characteristics $(\tau_b, \theta_b, \tau_{eq}, \Psi_{eq}, \Delta_{eq})$ of the two matrices $\mathbf{P}_b$ and $\mathbf{R}_{eq}$ can be experimentally determined, as in Eqs. (25), from the eigenvalues of $(\mathbf{a}\mathbf{r}_1\mathbf{w})^{-1}(\mathbf{a}\mathbf{r}_2\mathbf{w})$ and the trace of $(\mathbf{a}\mathbf{r}_1\mathbf{w})^{-1}(\mathbf{a}\mathbf{r}_1\mathbf{p}_b\mathbf{w})$.

The characteristics of $(\mathbf{R}_1, \mathbf{R}_2)$ and the orientation of $\mathbf{P}_b$ must be chosen such that $\mathbf{R}_{eq}$ and $\mathbf{P}_b$ fulfill conditions (24) and that $\mathbf{R}_1$ is invertible ($|\sin 2\Psi_1| \gg 0$).

The value of matrix $\mathbf{A}$ cannot be determined directly from $\mathbf{W}$ as in the first case because the product $(\mathbf{a}\mathbf{w})$ is no longer available. Moreover, it cannot be obtained from either $(\mathbf{a}\mathbf{r}_1\mathbf{w})$ or $(\mathbf{a}\mathbf{r}_2\mathbf{w})$ because the experimental characteristics of $\mathbf{R}_1$ and $\mathbf{R}_2$ are unknown. As a consequence, $\mathbf{A}$ is calibrated like $\mathbf{W}$ by use of the fourth measurement $(\mathbf{a}\mathbf{p}_a\mathbf{r}_1\mathbf{w})$: It is in the

null space of

$$\begin{aligned}\mathbb{H}_{\mathbf{P}_a}:\mathbf{X} &\rightarrow \mathbf{X}\mathbf{P}_a - (\mathbf{a}\mathbf{p}_a\mathbf{r}_1\mathbf{w})(\mathbf{a}\mathbf{r}_1\mathbf{w})^{-1}\mathbf{X}, \\ \mathbb{H}'_{\mathbf{R}_{eq}}:\mathbf{X} &\rightarrow \mathbf{X}\mathbf{R}_{eq} - (\mathbf{a}\mathbf{r}_2\mathbf{w})(\mathbf{a}\mathbf{r}_1\mathbf{w})^{-1}\mathbf{X}.\end{aligned}\quad (30)$$

This system of equations is solved as in the previous case, and the vector representation $\mathbf{A}^{16}$ of $\mathbf{A}$ is found to be the eigenvector of $\mathbf{K}'$ associated with its smallest quasi-null eigenvalue λ_{16}' , as follows:

$$\mathbf{A}^{16} = {}^T\mathbf{O}' \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}, \quad (31)$$

where

$$\begin{aligned}\mathbf{K}' &= {}^T\mathbf{H}_{\mathbf{P}_a}\mathbf{H}_{\mathbf{P}_a} + {}^T\mathbf{H}_{\mathbf{R}_{eq}}\mathbf{H}_{\mathbf{R}_{eq}} \\ &= {}^T\mathbf{O}' \begin{pmatrix} \lambda_1' & & 0 \\ & \ddots & \\ 0 & & \lambda_{16}' \end{pmatrix} \mathbf{O}' \\ \lambda_1' &> \dots > \lambda_{15}' \gg \lambda_{16}' \approx 0.\end{aligned}\quad (32)$$

E. Error Control

1. Definition

The relative errors ε_W and ε_A in $\mathbf{W}$ and $\mathbf{A}$, which are due to experimental errors, are defined by

$$\varepsilon_W = \frac{|\mathbf{W} - \mathbf{W}_{ECM}|}{|\mathbf{W}|}, \quad \varepsilon_A = \frac{|\mathbf{A} - \mathbf{A}_{ECM}|}{|\mathbf{A}|}, \quad (33)$$

where

$$|\mathbf{M}| = \left(\sum_{i,j} M_{ij}^2 \right)^{1/2} \quad (34)$$

is the Frobenius norm, $\mathbf{A}$ and $\mathbf{W}$ are the actual matrix values, and $\mathbf{A}_{ECM}$ and $\mathbf{W}_{ECM}$ are those values determined by the ECM. Usually these calibration errors are unknown because of the lack of knowledge of the values of $\mathbf{A}$ and $\mathbf{W}$. Nevertheless, one of the main advantages of the ECM resides in its capability to estimate those errors by use of error estimators. Indeed, the reference samples were chosen such that the null spaces of matrices $\mathbf{K}$ and $\mathbf{K}'$ contain only the matrices $\mathbf{W}$ and $\mathbf{A}$ (see Subsection 3B.1). Consequently $\mathbf{K}$ and $\mathbf{K}'$ should have one, and only one, null eigenvalue. Yet, because of experimental errors, one can find only small eigenvalues. Consequently it is possible to control the influence of errors with the error estimators ε_{ECM} and ε'_{ECM} that reflect the contrast between the null and the nonnull eigenvalues:

$$\varepsilon_{ECM} = \sqrt{\lambda_{16}/\lambda_{15}}, \quad \varepsilon'_{ECM} = \sqrt{\lambda'_{16}/\lambda'_{15}}. \quad (35)$$

The mathematical connection between the actual relative errors ε_W and ε_A and their estimators ε_{ECM} and ε'_{ECM} is complex and beyond the scope of this paper. On the other hand, a numerical simulation

Table 2. Measured and Theoretical Values of the Characteristics of the Reference Samples at $\lambda = 458$ nm and $\lambda = 633$ nm

Characteristic	458 nm		633 nm	
	Measured	Theoretical	Measured	Theoretical
τ_{eq}	0.565	0.546	0.435	0.436
Ψ_{eq}	30.69°	30.19°	24.56°	24.30°
Δ_{eq}	51.84°	49.9°	40.87°	39.78°
τ_b	0.837	—	0.847	—
τ_a	0.828	—	0.824	—

will efficiently demonstrate the relevance of these estimators.

2. Numerical Simulation

The numerical simulation uses the measured values **A** and **W** of our polarimeter and PSG and the theoretical Mueller matrices **P_b**, **P_a**, **R₁**, and **R₂** of the samples, obtained from definitions (2) and (5) and Table 2, to simulate the measured matrices (**amw**) = **AMW**. Then we take the experimental errors into account by adding random noise with a controlled amplitude $|\Delta(\mathbf{amw})| = 0.005|\mathbf{amw}|$ to these matrices. Finally, the ECM is implemented, which provides the calibration values **A_{ECM}** and **W_{ECM}** and the error estimators ϵ_{ECM} and ϵ'_{ECM} . In that case, as the actual values **A** and **W** are known, calibration errors ϵ_A and ϵ_W can be estimated and compared with the error estimators.

Figure 2 displays the calibration errors ϵ_A and ϵ_W for 50 calibrations with the same noise amplitude (0.5%). The variations are due to the random structure of the noise matrices. The average errors $\langle\epsilon_A\rangle = 0.25$ and $\langle\epsilon_W\rangle = 0.29$ are two times less than the injected noise, which demonstrates that the ECM is an accurate and robust calibration method. Figure 3 displays the ratio between the actual and the estimated errors. The values of the average ratios $\langle\epsilon_A/\epsilon'_{ECM}\rangle$

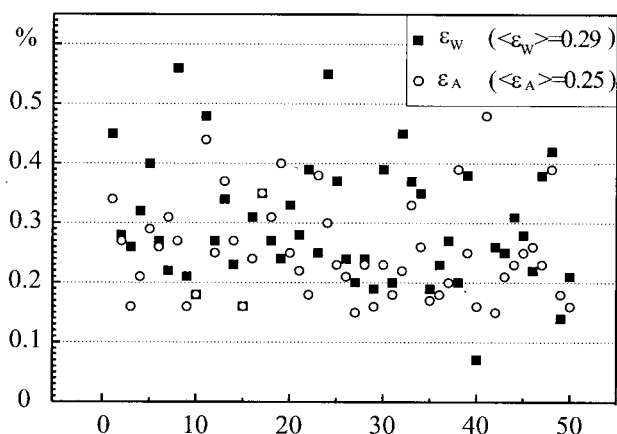


Fig. 2. Fifty numerical simulations of the calibration accuracy. The measurement errors are simulated by addition of random matrices to the theoretical measurements. Relative errors in the measurements are taken to be equal to 0.5% and result, with the ECM, in average relative error $\langle\epsilon_A\rangle = 0.25\%$ in **A** and $\epsilon_W = 0.29\%$ in **W**.

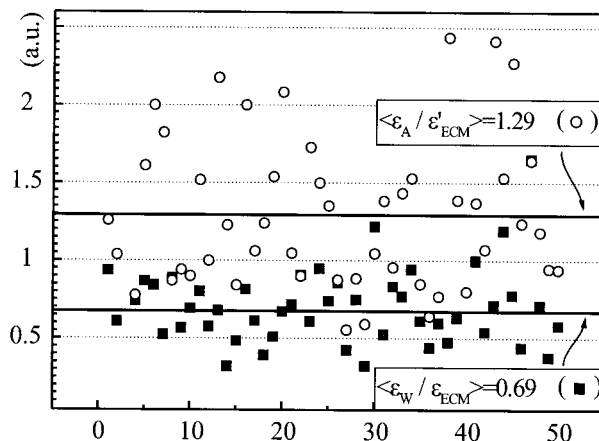


Fig. 3. Fifty numerical simulations of the ratio between the actual calibration errors (ϵ_A , ϵ_W) in (**A**, **W**) and the error estimators ϵ_{ECM} and ϵ'_{ECM} .

$\epsilon'_{ECM}\rangle = 0.69$ and $\langle\epsilon_W/\epsilon_{ECM}\rangle = 1.29$ together with the limited data dispersion demonstrate that the error estimators are meaningful. In using the ECM, one gets the values of the calibration matrices **W** and **A**, which is the primary reason for using the ECM, and also a confidence interval.

F. Calibration Algorithm

The overall calibration procedure is schematically summarized in Fig. 4.

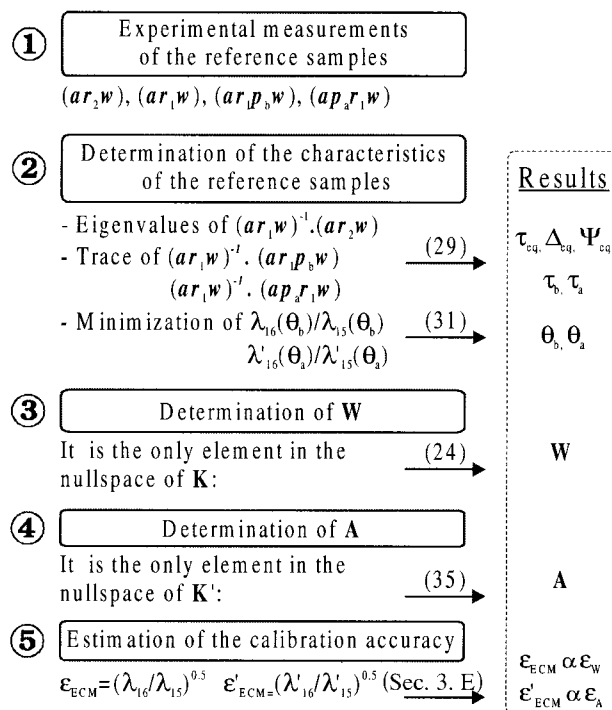


Fig. 4. ECM algorithm. Numbers in parentheses refer to the corresponding equations in text.

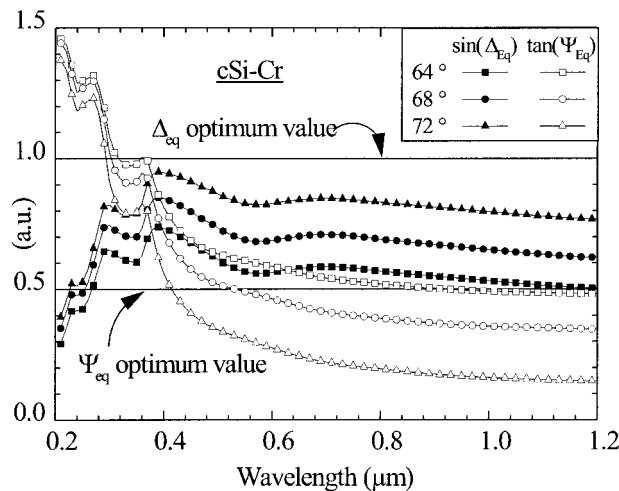


Fig. 5. Equivalent polarimetric properties of *c*Si-Cr samples. The values are plotted as a function of the wavelength for several angles of incidence.

4. Experimental Results

The ECM procedure was implemented to calibrate a multiwavelength MME.⁶ We first used a configuration compatible with transmission and reflection and then a configuration restricted to reflection, which is the present geometric configuration of the MME. The presentation of experimental results obtained with our MME at two wavelengths (458 and 633 nm) by use of the reflection-limited configuration will illustrate the effective efficiency and simplicity of using the ECM.

A. Choice of the Calibration Samples

The two polarizers used in this experimental exercise are Glan prisms. They have extinction coefficients smaller than 10^{-5} and can be considered perfect with respect to commonly available precision.

The two reference samples R_1 and R_2 are chosen such that R_{eq} , defined by Eq. (29), has overall characteristics close to the optimum values given in Eq. (24). These characteristics depend on the wavelength and the angle of incidence. As a consequence, it is difficult to find universal samples that can be used in any configuration. Nevertheless, many samples, eventually coated, can be used to fit any special configuration. Moreover, their flexibility and simplicity of use are considerably improved by the self-consistency of the ECM: The characteristics of the samples merely have to be roughly close to the optimum to minimize error propagation; their exact values are determined during the calibration. Satisfying properties are obtained by use of a common *c*-Si wafer associated with either chromium- or palladium-coated glass. The dispersion of the theoretical polarimetric properties Ψ_{eq} and Δ_{eq} , defined in Eq. (29), are plotted in Figs. 5 and 6 for the *c*Si-Cr and the *c*Si-Pd samples, respectively. Displaying a small spectral dispersion²⁵ for wavelengths under the optical gap of silicon, they provide satisfying properties in a broad spectrum (at least 400–800 nm) and

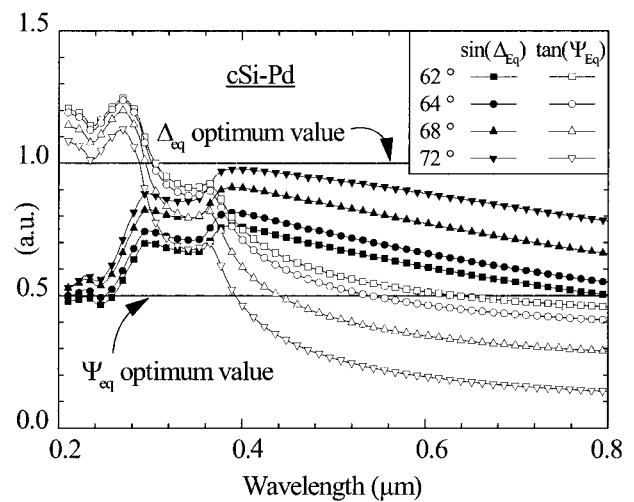


Fig. 6. Equivalent polarimetric properties of *c*Si-Pd samples. The values are plotted as a function of the wavelength for several angles of incidence.

for a convenient range of angles of incidence (64° – 70°).

Finally, the best experimental results presented in this paper were obtained with *c*-Si- and Pd-coated glass. As a matter of fact, the results of the calibration with Cr mirrors were altered by stresses in the deposited layer that induced perturbations in the optical response of this material. One of the solutions, to prevent surface anisotropy from inducing cross-polarization effects, lies in using uncoated bulk materials, which would limit surface stresses. The calibration results presented in this paper were recorded at an angle of incidence of 64° . R_1 and R_2 correspond, respectively, to reflection on the Pd and on the *c*-Si sample.

B. Samples Characteristics

The characteristics of the reference samples are determined from Eqs. (25): The equivalent sample matrix $R_{eq} = R_1^{-1}R_2$ from the eigenvalues of the measured matrices $(ar_1w)^{-1}(ar_2w)$ and the transmission coefficients of the polarizers from the traces of $(ar_1w)^{-1}(ar_1p_bw)$ and $(ar_1w)^{-1}(ap_ar_1w)$. These values are compared in Table 2 with theoretical ones obtained from Ref. 25 and Eq. (29). The theoretical values do not take into account the thickness of the Pd layer and the native oxide layers and possible roughness on the top of the reference samples. This may explain the small discrepancy, without practical consequences, between experimental and theoretical values. The transmission coefficients of the polarizers increase with the wavelength because of the increase of the refractive index.

The geometrical orientations of the entrance and exit polarizers are determined from expression (27). Figure 7 shows that the geometric orientations of the polarizers are easily determined without any ambiguity. One gets $\theta_a = 46.75^\circ$ for the polarizer introduced between the sample and the polarimeter (ap_ar_1w) and $\theta_b = 45.80^\circ$ for the polarizer introduced

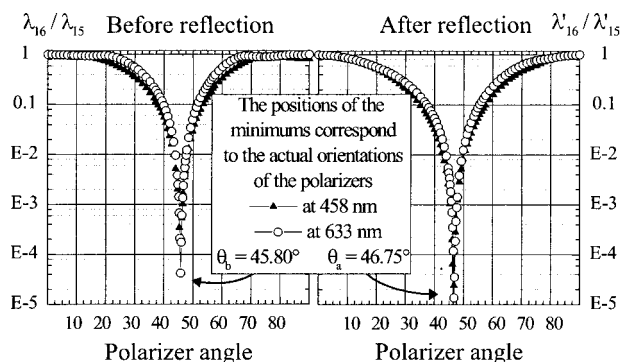


Fig. 7. Determination of the orientations of the two polarizers. The correct orientations θ_b and θ_a minimize the ratios $\lambda_{16}/\lambda_{15}$ and $\lambda'_{16}/\lambda'_{15}$.

between the sample and the PSG ($\mathbf{ar}_1\mathbf{p}_b\mathbf{w}$). As expected, the orientations of the polarizers are found to be independent of the wavelength. This experimental determination is important on the one hand because it increases the calibration precision and on the other hand because it obviates the need to use expensive high-precision rotating holders to orient the polarizers accurately. The polarizers merely need to be roughly oriented at $\sim 45^\circ$.

C. Error Control

The overdetermination of calibration matrices $\mathbf{W}$ and $\mathbf{A}$ allows us to get extra information. We have already explained how the characteristics of the samples are determined from eigenvalues of the measured matrices. In the same manner, the study of the eigenvalues of $\mathbf{K}$ and $\mathbf{K}'$ provides an important estimation of the final accuracy of the calibration (see Subsection 3.E).

The eigenvalues of $\mathbf{K}$ and $\mathbf{K}'$ are displayed in Fig. 8. The control of their amplitudes is important for two reasons. The first is that the accuracy of the calibration is affected by the balance among the non-null values (see Subsection 3.B.2). A lack of balance would increase the error transmission, and a well-

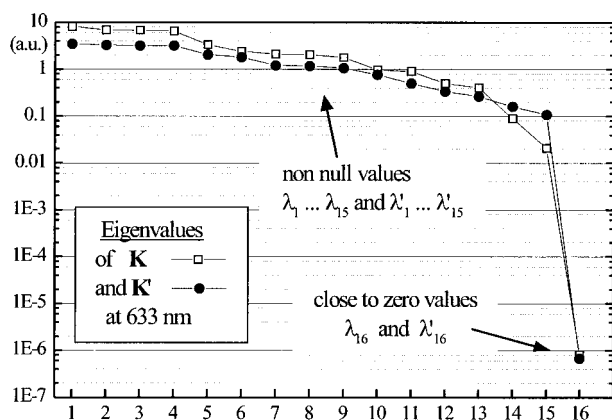


Fig. 8. $\mathbf{K}$ and $\mathbf{K}'$ eigenvalues at 633 nm. They reflect the behavior of the calibration procedure ruled by the choice of the reference samples and the accuracy of the basic measurements.

balanced calibration is obtained by use of samples close to the optimum values displayed in Eq. (24). The second reason is related to the necessity to provide the values of the error estimators ϵ_{ECM} and ϵ'_{ECM} defined in Eqs. (35):

$$\begin{aligned}\epsilon_{\text{ECM}} &= 0.34\%, \\ \epsilon'_{\text{ECM}} &= 0.98\%.\end{aligned}\quad (36)$$

These estimators are affected less by the choice of the calibration samples than by the accuracy of the basic measurements. Consequently this contrast reflects the ability to provide accurate measurements. A contrast of several orders of magnitude as in the present case proves that the behavior of the MME is satisfying. By using the estimator values and the results of the numerical simulation in Subsection 3.E one can estimate that the calibration error does not exceed 0.5%.

D. Accuracy of the Measurements

We evaluated the accuracy of this calibration procedure by measuring the Mueller matrix of a SiO_2 layer deposited by plasma-enhanced chemical-vapor deposition²⁶ upon a monocrystalline (111) silicon wafer cleaned by standard procedure.²⁷ The layer thickness estimated from *in situ* ultraviolet-visible spectroscopic (1.5–5-eV) phase-modulated ellipsometry is $1.04 \pm 0.005 \mu\text{m}$, and the refractive index is close to those of thermal silica, 1.465 ± 0.005 (at 458 nm) and 1.457 ± 0.005 (at 633 nm). Figure 9 displays the variations of the 16 Mueller-matrix coefficients as a function of the angle of incidence for two wavelengths (458 and 633 nm). The experimental values are compared with the theoretical ones; the thickness of the SiO_2 layer is fitted. The estimated thickness is $1.0408 \pm 0.001 \mu\text{m}$ (at 458 nm) and $1.0409 \pm 0.001 \mu\text{m}$ (at 633 nm), which corresponds to the phase-modulated ellipsometry value. Finally, except for the grazing incidences ($\phi_i > 85^\circ$) for which the laser beam section is larger than the sample, the maximum relative error for the 16 Mueller-matrix coefficients is also found to be less than 0.5%.

5. Summary and Conclusions

A new calibration procedure, the eigenvalue calibration method, has been presented. It is devoted to the complex and critical calibration of polarization-state generators, polarimeters, and Mueller matrix ellipsometers. An optimized universal procedure based on the extensive use of matrix algebra has been described. It relies on the measurements of three or four reference samples whose few characteristics are determined during the calibration. It strongly facilitates the implementation of a MME without the need for precise orientation of the various elements. The numerous inherent defects of the optical elements that constitute the polarization-state generator and the polarimeter, including eventual filters, beam expanders, windows, or mirrors, are automatically taken into account by the ECM. Furthermore,

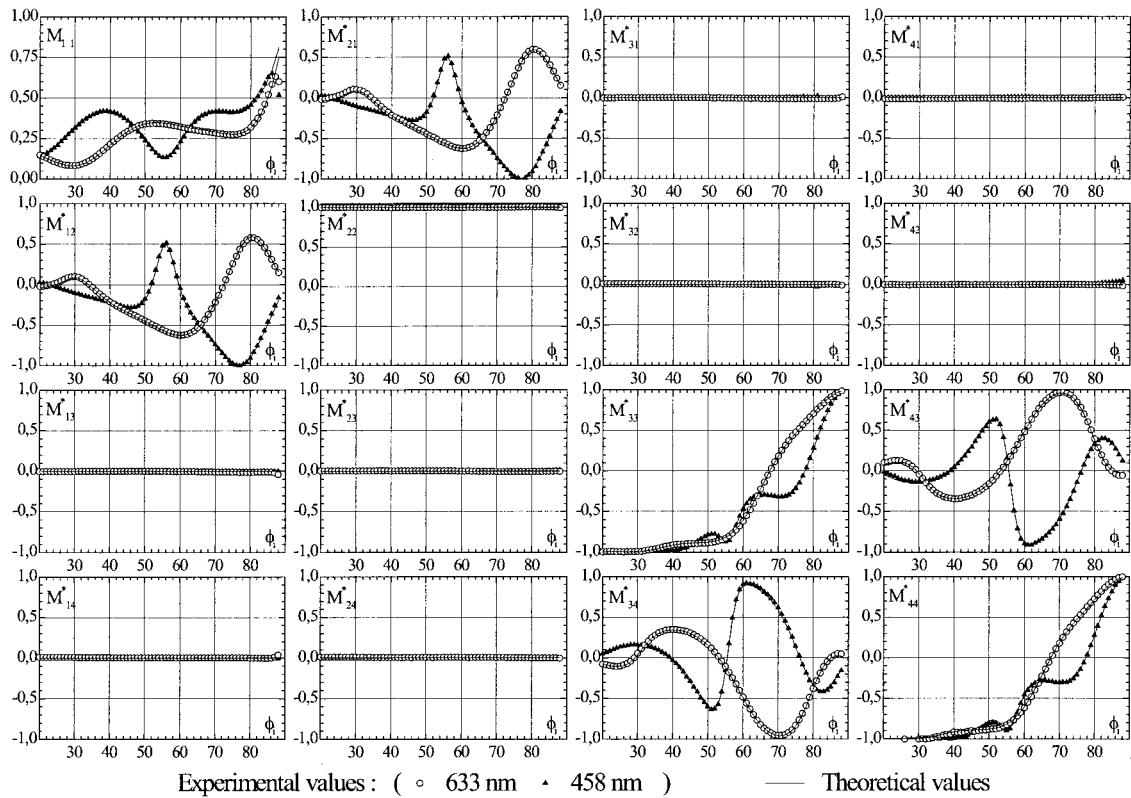


Fig. 9. Measured Mueller matrices of a SiO₂-coated *c*-Si substrate as a function of angle of incidence (ϕ_i) for two wavelengths (458 and 633 nm). The coefficient M_{11} is the intensity reflectance, and the other coefficients, normalized by M_{11} , display the complete polarimetric response. The error between experimental and theoretical values (solid curves) is less than 0.5%.

a meaningful error-control indicator evaluates the accuracy of the calibration.

Until now the optical characterization of scattering samples has been limited by difficulties in carrying out MME measurements. It can be expected that the ECM will contribute to overcoming this experimental limitation and enable MME users to focus on the physical interpretation of the measurements.

Appendix A: Choice of the Reference Sample

The proof that the choice of the reference sample does not depend on $\mathbf{W}$ and $\mathbf{A}$, which are invertible matrices, is quite direct: Considering the change $\mathbf{X}' = \mathbf{X}\mathbf{W}^{-1}$ in the variables of the mapping $\mathbb{H}_M$, defined in relation (14), neglecting the experimental errors $(\mathbf{a}\mathbf{w})^{-1}(\mathbf{a}\mathbf{m}\mathbf{w}) \rightarrow \mathbf{W}^{-1}\mathbf{M}\mathbf{W}$, and noting that the mappings $\mathbf{X} \rightarrow \mathbb{H}_M(\mathbf{X})$ and $\mathbf{X} \rightarrow \mathbb{H}_M(\mathbf{X})\mathbf{W}^{-1}$ have the same null space, one gets

$$\mathbb{H}_M(\mathbf{X}) = 0 \Leftrightarrow [\mathbf{M}, \mathbf{X}'] = 0, \quad (\text{A1})$$

where $[\mathbf{M}, \cdot]$ designates the Lie product or commutator of $\mathbf{M}$, which is also a mapping, like $\mathbb{H}_M$, from the set $\mathbb{M}_4(\mathbb{R})$ into itself:

$$[\mathbf{M}, \cdot]: \mathbb{M}_4(\mathbb{R}) \rightarrow \mathbb{M}_4(\mathbb{R}), \quad \mathbf{X}' \rightarrow \mathbf{M}\mathbf{X}' - \mathbf{X}'\mathbf{M}. \quad (\text{A2})$$

Relation (A1) shows that the solutions of Eq. (15) are influenced only by matrices $\mathbf{M}$, independently of $\mathbf{A}$ and $\mathbf{W}$.

Transforming the calibration problem in this manner has two advantages: The first is to get rid of the apparent dependence on $\mathbf{A}$ and $\mathbf{W}$; the second is to get a well-established, extensively studied mathematical formulation. As a matter of fact, $\mathbf{W}$ will be unambiguously determined by Eq. (15) if the only matrix $\mathbf{X}'$ that verifies

$$[\mathbf{M}, \mathbf{X}'] = 0, \quad \forall \mathbf{M} \in \{\mathbf{M}\} \quad (\text{A3})$$

is the identity matrix $\mathbf{I}_4$; then $\mathbf{X}' = \mathbf{I}_4$ implies that $\mathbf{X} = \mathbf{W}$.

Appendix B. Size of $\mathbb{H}_M$ Null Space: Reference Sample Eigenvalues

Relation (A1) shows that the null space of $\mathbb{H}_M$ and the commutator of $\mathbf{M}$ have the same dimension. In the present case, in which the reference matrices defined in Eqs. (3) and (5) are 4×4 diagonalizable matrices, the dimension of the vector space that commutes with them is a function of the multiplicity of their eigenvalues. As a matter of fact, if $P(x)$ is the characteristic polynomial of $\mathbf{M}$,

$$P(x) = \det(x\mathbf{I}_4 - \mathbf{M}) = \prod_{i=1 \dots m} (x - \lambda_i)^{\alpha_i}, \quad (\text{B1})$$

m is the number of eigenvalues λ_i , and α_i is their multiplicity, one can easily demonstrate (by working

in a diagonalization basis of $\mathbf{M}$) that the dimension of the commutator null space is

$$\sum_{i=1 \dots m} \alpha_i^2. \quad (\text{B2})$$

Consequently the null space of $\mathbb{H}_P$ contains 10 solutions because $\mathbf{P}(\tau_p, \theta)$, defined in Eq. (3), has two distinct eigenvalues:

$$\tau_p \text{ (multiplicity, 1),} \quad 0 \text{ (multiplicity, 3)} \quad (\text{B3})$$

and $\mathbb{H}_R$ contains four solutions because $\mathbf{R}(\tau_R, \Psi, \Delta)$, defined in Eq. (5), has two real and two complex distinct eigenvalues:

$$\begin{aligned} 2\tau_R \sin^2(\Psi), & \quad 2\tau_R \cos^2(\Psi); \\ \tau_R \sin(2\Psi)\exp(i\Delta), & \quad \tau_R \sin(2\Psi)\exp(-i\Delta). \end{aligned} \quad (\text{B4})$$

Appendix C. Matrix $\mathbf{K}$ Conditioning

The basic necessary condition for matrix $\mathbf{K}$, defined in Eq. (20), is to have only one null, nondegenerated eigenvalue, which implies that

$$\lambda_{16}/\lambda_{15} \ll 1. \quad (\text{C1})$$

Without noise, this condition is always fulfilled because $\lambda_{16} = 0$. Without trying to give a general demonstration, we illustrate by a particular case that a condition to satisfy inequality (C1) when experimental errors occur is

$$\lambda_{15}/\lambda_1 \gg \text{error level}. \quad (\text{C2})$$

We consider that the experimental error induces a relative perturbation on $\mathbf{K}$ less than ε :

$$|\Delta\mathbf{K}|/|\mathbf{K}| \leq \varepsilon, \quad (\text{C3})$$

and note, using Eq. (20) and the definition of the norm $||$ in (34), that

$$\lambda_1 \leq |\mathbf{K}| \leq 4\lambda_1. \quad (\text{C4})$$

Then, considering a particular perturbation $\Delta\mathbf{K}$ that is proportional to identity matrix $\mathbf{I}_{16}$ to commute with $\mathbf{K}$ while verifying inequality (C3),

$$\begin{aligned} \mathbf{K} + \Delta\mathbf{K} &= \mathbf{K} + \frac{\varepsilon}{4} \lambda_1 \mathbf{I}_{16} \\ &= {}^t\mathbf{O} \begin{bmatrix} \lambda_1 + \frac{\varepsilon}{4} \lambda_1 & & & \\ & \ddots & & \\ & & \lambda_{15} + \frac{\varepsilon}{4} \lambda_1 & \\ & & & \ddots & \\ & & & & \frac{\varepsilon}{4} \lambda_1 \end{bmatrix} \mathbf{O}, \end{aligned} \quad (\text{C5})$$

one obtains for condition (C1)

$$\lambda_{15}/\lambda_1 \gg \varepsilon/4. \quad (\text{C6})$$

We have verified that a necessary condition is that the eigenvalues of $\mathbf{K}$ have to be well balanced.

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